

Centrifugal Pump Manual — Model P-100 Series (incl. P-102)

1. General Description

The P-100 series is a horizontal centrifugal pump used for transferring process water and light hydrocarbons in industrial plants. Unit P-102 is installed in the cooling water loop of Process Line 2 and operates continuously (24/7) under normal plant conditions.

2. Technical Specifications

Rated flow: 250 m³/h. Rated head: 45 m. Motor power: 75 kW. Maximum allowed bearing temperature: 85 degrees Celsius. Maximum allowed casing temperature: 90 degrees Celsius. Lubrication: oil bath, ISO VG 68.

3. Cooling and Lubrication System

The pump bearings are cooled by an auxiliary cooling water jacket fed from the same cooling loop the pump serves. If the cooling water supply pressure drops below 1.5 bar, bearing cooling becomes insufficient and bearing temperature will rise even if the pump itself is operating within its normal flow range.

4. Common Causes of Overheating

Overheating on P-100 series pumps, including P-102, is most frequently caused by one or more of the following:

- Insufficient lubrication oil level or degraded oil (check sight glass and oil color).
- Blocked or restricted cooling water jacket, reducing bearing cooling capacity.
- Pump operating away from Best Efficiency Point (BEP), i.e. at very low flow (below 30% of rated flow) or very high flow, which increases internal recirculation and heat generation.
- Misalignment between pump and motor shaft, causing excessive bearing friction.
- Worn or damaged bearings requiring replacement.
- Cavitation due to insufficient suction pressure (NPSH available lower than NPSH required).

5. Recommended Inspection Sequence

When P-102 (or any pump in this series) shows high temperature alarms, technicians should follow this sequence before disassembly:

- Verify oil level and oil condition at the sight glass.
- Check cooling water supply pressure at the jacket inlet (minimum 1.5 bar required).
- Confirm the pump is operating within 70-110% of the Best Efficiency Point flow.
- Check shaft alignment with dial indicator or laser alignment tool.
- Listen/measure vibration at bearing housings; compare against baseline vibration signature.
- If all above are normal, schedule bearing inspection at next planned shutdown.